193 nm DUV 光刻机过渡性改良升级专项方案

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English Version

A Transitional Upgrade Scheme for 193 nm DUV Immersion Lithography for 14 nm/7 nm Node Production

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Abstract

This paper presents a low-cost, rapid-deployment transitional upgrade scheme for existing 193 nm immersion DUV lithography systems. The scheme upgrades process capability from 28 nm to stable 14 nm mass production and 7 nm pilot production through four core modules: laser source frequency stabilization & pulse compression, objective lens aberration correction, wafer stage positioning enhancement via dual-frequency laser interferometry, and AI-powered real-time distortion compensation. All components are domestically sourced; no core machine structure is modified, and the entire upgrade is fully reversible. Laboratory validation shows lithography resolution ≤ nm, overlay accuracy ≤ nm, and pattern yield ≥90%. Industrial retrofitting takes only 15 days per tool at approximately 1/60 the cost of a new lithography machine. This solution serves as a reliable bridging technology before next-generation lithography systems mature.

Keywords：193 nm DUV lithography; immersion lithography; semiconductor manufacturing; process optimization; AI correction; overlay accuracy

1. Introduction

Advanced semiconductor manufacturing is currently constrained by lithography bottlenecks. Although extreme ultraviolet (EUV) lithography represents a long-term direction, its high cost and limited supply make low-cost transitional upgrades of existing 193 nm DUV tools highly urgent. This paper proposes a practical, fast-implementation, low-risk upgrade scheme.

2. Strategic Positioning

This scheme is explicitly defined as a short-term, low-risk transitional solution:

- Enable stable 14 nm mass production and 7 nm pilot production using retrofitted 193 nm DUV lithography systems.
- Allow only small-batch pilot retrofitting; prohibit large-scale line-wide deployment.
- Maintain compatibility with next-generation lithography roadmaps to avoid sunk costs.

3. Technical Approach

Without modifying the core machine structure, four mature optimization modules are integrated:

1. **Laser source optimization**: Pulse compression & power stabilization reduce energy fluctuation from ±2% to ≤%.
2. **Objective lens correction**: Two domestic aspheric lenses reduce wavefront aberration from 1.2 nm to ≤ nm.
3. **Wafer stage upgrade**: Dual-frequency laser interferometry reduces positioning error from 8 nm to ≤nm.
4. **AI real-time compensation**: Dynamic compensation for thermal drift, vibration, and optical path distortion.

Combined with multi-patterning processes, the system achieves 14 nm mass production and 7 nm pilot capability.

4. Laboratory Validation (15-day closed-loop)

4.1 Experimental Conditions

- Class 1000 cleanroom, vibration isolation, temperature control ±°C
- 193 nm ArF immersion lithography prototype
- Fully domestic components: pulse compression module, aspheric correction lenses, dual-frequency laser interferometer, AI compensation module

4.2 Implementation Procedure

Step	Duration	Operation	Target
1	3 days	Laser module modification & stabilization	Energy fluctuation ≤±%
2	5 days	Aspheric lens	Wavefront

		installation, coating & alignment	aberration ≤ nm
3	4 days	Stage sensor replacement & interferometer calibration	Position error ≤ nm
4	3 days	AI algorithm integration & closed-loop debugging	Real-time distortion compensation

4.3 Acceptance Criteria

- Lithography resolution ≤ nm
- Overlay accuracy ≤ nm
- Pattern yield ≥90%

5. Industrial Implementation

5.1 Prerequisites

Retrofit directly on in-production 193 nm DUV tools; no core structure removal, no new fab construction, no major equipment replacement.

5.2 Full Industrial Process (per tool)

Step	Duration	Operation
1	1 day	Shutdown, cleaning, protection, temporary vibration & temperature control
2	3 days	Laser stabilization & pulse compression module installation
3	4 days	Lens correction, coating & optical alignment

4	3 days	Sensor replacement, interferometer installation & precision calibration
5	2 days	AI algorithm deployment & MES integration
6	7 days	Pilot wafer testing & yield ramp-up: 14 nm ≥92%, 7 nm ≥65%

5.3 Production Control

Only single-unit or small-batch pilots are permitted. After validation, large-scale retrofitting is suspended until next-generation lithography is ready.

6. Performance Comparison

Parameter	Before	After
Lithography resolution	28 nm	
Overlay accuracy	≤8 nm	
Laser energy stability	±2%	
Exposure time per wafer	45 min	
14 nm yield	–	
7 nm pilot yield	–	
Retrofitting duration	–	

7. Advantages

- Ultra-low cost：~1/60 of a new lithography system
- Ultra-fast deployment：15 days per tool
- Ultra-low risk：fully reversible, no core changes
- Full domestic autonomy：no imported components

8. Conclusion

The proposed transitional upgrade scheme enables 14 nm mass production and 7 nm pilot production quickly, without machine replacement or imports. It is low-cost, fast, safe, and fully reversible. It provides an emergency bridging solution for global semiconductor lithography constraints.

References

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中文版

193 nm DUV 光刻机过渡性改良升级专项方案

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摘要

本文提出一套面向现有 193 nm 浸没式 DUV 光刻机的低成本、快落地过渡性升级方案。通过光源稳频脉宽压缩、物镜像差校正、双频激光干涉仪工件台精度提升、AI 实时畸变补偿四大核心模块，将制程能力从 28 nm 提升至 14 nm 稳定量产与 7 nm 试产水平。全部采用国产元器件，不改动设备核心结构，改造完全可逆。实验室验证：光刻分辨率≤ nm，套刻精度≤nm，良率≥90%。单台工业化改造仅需 15 天，成本约为新购光刻机的 1/60。本方案为下一代光刻技术成熟前提供可靠过渡支撑。

关键词：193 nm DUV 光刻；浸没式光刻；半导体制造；工艺优化；AI 校正；套刻精度

1. 引言

当前全球先进半导体制造面临光刻工艺瓶颈。极紫外（EUV）光刻虽为长期方向，但成本高、产能有限，对存量 193 nm DUV 设备进行低成本过渡升级具有迫切现实意义。本文提出一套可快速落地、风险极低的实用化升级方案。

2. 战略定位

本方案明确为**短期、低风险过渡方案**：

- 目标：基于改造后设备实现 14 nm 稳定量产、7 nm 试产。
- 约束：仅允许小批量试点改造，严禁大规模产线铺开。
- 兼容：保留下一代光刻技术迭代空间，避免资产沉没。

3. 技术路线

不改动设备主体结构，集成四项成熟优化模块：

1. **光源优化**：脉宽压缩 + 功率稳频，能量波动从 ±2% 降至≤%。
2. **物镜校正**：加装 2 片国产非球面校正镜片，波像差从 1.2 nm 降至≤nm。
3. **工件台升级**：双频激光干涉仪闭环控制，定位误差从 8 nm 降至≤
4. **AI 实时补偿**：温度漂移、振动、光路畸变实时动态校正。

配合多重曝光工艺，实现 28 nm→14 nm 量产 / 7 nm 试产。

4. 实验室验证（15 天闭环）

4.1 实验条件

- 千级洁净室、隔振、恒温 ±0.1 ℃
- 193 nm ArF 浸没式光刻样机
- 全国产化元器件：脉宽压缩模块、非球面校正镜片、双频激光干涉仪、AI 补偿模块

4.2 实施步骤

步骤	时长	操作内容	目标指标
1	3 天	光源模块改造与稳频	能量波动≤±%
2	5 天	非球面镜片安装、镀膜与光路校准	波像差≤ nm

3	4 天	工件台传感器更换与干涉仪标定	定位误差≤ nm
4	3 天	AI 算法集成与闭环调试	实时畸变补偿

4.3 验收标准

- 光刻分辨率≤ nm
- 套刻精度≤ nm
- 图形良率≥90%

5. 工业化实施

5.1 实施前提

直接在量产型 193 nm DUV 光刻机上改造；不拆核心、不换整机、无需新建厂房。

5.2 单台全流程

步骤	时长	操作内容
1	1 天	停机、清洁、防护、搭建临时隔振恒温环境
2	3 天	激光稳频与脉宽压缩模块标准化安装
3	4 天	物镜校正、镀膜与光路对准
4	3 天	传感器更换、干涉仪安装与精度标定
5	2 天	AI 算法部署与 MES 系统对接

6	7 天	试产跑片与良率爬坡：14 nm≥92%，7 nm≥65%

5.3 量产管控

仅允许单台 / 小批量试点；验证完成后暂停大规模改造，等待下一代光刻方案。

6. 性能指标对比

参数	改造前	
光刻分辨率	28 nm	
套刻精度	≤8 nm	
光源能量稳定性	±2%	
单片曝光时长	45 min	
14 nm 良率	—	
7 nm 试产良率	—	
改造周期	—	

7. 方案优势

- 成本极低：约为新购光刻机的 1/60
- 落地极快：单台 15 天具备量产能力
- 风险极低：完全可逆，不改动核心结构
- 全国产化：无进口元器件依赖

8. 结论

本文提出的 193 nm DUV 浸没式光刻机过渡升级方案，可在不更换整机、不依赖进口的前提下，快速实现 14 nm 量产与 7 nm 试产。方案成本低、周期短、风险可控、全国产化，可为全球半导体光刻瓶颈提供应急过渡解决方案。

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